

## Education

### Georgetown University

M.S. Computational Linguistics. GPA 4.0

Master's Research Paper: Investigating the Cross-Lingual Transferability of NLP Tasks

Washington, DC

2023-2026

### Dartmouth College

B.A. Major: Computer Science, Minor: Linguistics. GPA 3.74, Cum Laude

Hanover, NH

2017-2021

## Experience

### Johns Hopkins University Applied Physics Laboratory

Laurel, MD

#### Natural Language Processing Researcher / Software Engineer

Aug 2021 – Present

- Created software for U.S. Government that inject domain knowledge into language models and extract critical information from text using state of the art NLP methods and Domain Ontologies
- Awarded 8 grants to pursue research in AI transparency, concept erasure, summarization, entity disambiguation, LLM domain adaptation, AI red teaming, and multimodal anomaly detection
- Led research teams of 3-6 people, presented results to internal and external stakeholders

### DALI Lab

Hanover, NH

#### Software Developer

Jan 2019 – Jun 2021

- Produced full-stack web and mobile applications for external stakeholders every 10 weeks. Created and led classes on web design and development
- Developer mentor, Women in Science Program lead, core executive board member for 2020-2021

### Alarm.com

Tysons, VA

#### Software Engineer Intern

Jun 2020 – Aug 2020

- Created automated client-facing reports for property managers about all devices in their units

### Cambly Inc.

San Francisco, CA

#### Software Engineer Intern

Jan 2020 – Mar 2020

- Upgraded Android application to include English as a Second Language courses for adults.

### Dartmouth College Computer Science Department

Hanover, NH

#### Teaching Assistant

Sept 2018 – Dec 2018

- Taught weekly review class for Object Oriented Programming course to 10-15 students, graded coursework and tests

## Featured Research Projects

**Dandelion:** Identifying mission-relevant negative LLM behavior during runtime, steering in activation & weight space.

**Using Task Arithmetic for Cross-Lingual Transfer in LLMs: Meaning Transfers Better than Form:** Explores the cross-lingual transfer of tasks in LLMs as distance increases between source and target languages. (Paper under review)

**Task Arithmetic with Support Languages for Low-Resource ASR:** Incorporates information from genetically related high-resource languages to improve low-resource ASR models. [\[System Paper\]](#), to be published in an ACL workshop.

**CaTE:** Guidance on data curation methods for trustworthy AI development. [\[Full Report\]](#) [\[Paper on domain knowledge elicitation\]](#)

## Skills

**ML/NLP:** PyTorch, Tensorflow, Hugging Face, fine-tuning, LoRA, activation probing & steering, RAG, Slurm

**Software:** Python, JavaScript, TypeScript, React, Java, Kotlin, SQL, Elasticsearch, MongoDB, Git, Docker